

EXPERIENCE

Software Engineer

Amboss Technologies  [ambossTech](https://github.com/ambossTech)

Nov 2024 – Present

United States / Remote

- Own identity and paid access across Amboss products: built the OAuth 2.0 SSO platform with TOTP two-factor and the L402/LSAT pay-per-use API layer.
- Ship features end-to-end - NestJS GraphQL/REST APIs paired with React/Next.js frontends.
- Contribute across the Bitcoin/Lightning platform: Taproot Assets surfaces, Lightning liquidity products, and LND integrations.
- Deliver on AWS/Kubernetes with CI/CD pipelines and internal tooling.

Software Engineer / Maintainer

Openswap  [citadel-foss/openswap](https://github.com/citadel-foss/openswap)

Nov 2023 – Oct 2024

United States / Remote

- Maintained Openswap (formerly CoinSwap), a privacy-focused Bitcoin atomic-swap protocol, as a core Rust library maintainer for a year.
- Made wallets materially cheaper - binary CBOR storage cut wallet size by 28.8%.
- Redesigned protocol message encoding/decoding, speeding up integration tests by 21.7%.
- Modularized the Rust core - removed legacy paths and overhauled wallet configuration.
- Owned the project's public Rust API documentation.

Selected Fellow / Open Source SWE

Summer of Bitcoin  [citadel-foss/openswap](https://github.com/citadel-foss/openswap)

May 2024 – Sep 2024

United States / Remote

- Rebuilt the Openswap wallet from scratch on BDK's chain interfaces.
- Reimplemented WalletStore, Wallet, and public APIs around modular, maintainable storage.
- Added Fidelity and SwapCoin keychain variants plus RPC synchronization support.

Selected Fellow / Open Source SWE

Google Summer of Code  [xwiki-contrib/application-interactive-link-visualizer](https://github.com/xwiki-contrib/application-interactive-link-visualizer)

May 2023 – Sep 2023

France / Remote

- Built an installable graph explorer visualizing relationships between XWiki pages.
- Integrated XWiki APIs for backlinks, titles, URLs, and references, with search, facets, and theme-aware graph controls.
- Documented the extension publicly; produced the project blog post and demo video.

Freelance Software Engineer

Freelance  [wthrajat](https://github.com/wthrajat)

Dec 2021 – Apr 2023

India / Remote

- Freelanced via Upwork, Fiverr, and Freelancer, shipping products like GreatWall and iWriting.
- Built and maintained small web products, APIs, frontend surfaces, and automation workflows.
- Merged 2000+ pull requests across open-source and private work.

TECH SKILLS

Languages	TypeScript, Rust, Go, Python, JavaScript, C/C++, Solidity, Shell
Backend	Node.js, NestJS, gRPC, REST, GraphQL, PostgreSQL, Redis, OAuth 2.0, JWT
Frontend	React, Next.js, Astro.js, D3.js, Graphology, SigmaJS
DevOps	AWS, Docker, Kubernetes, CI/CD, Datadog, GitHub Actions
Blockchain	Bitcoin, Lightning Network, Taproot Assets, Payments, Wallets, Protocol testing
AI	MCP, RAG, LLM APIs, Claude Code, Codex
Styling	Tailwind CSS, Design systems

ACHIEVEMENTS

Summer of Bitcoin	selected from 5000+ applicants.
Linux Foundation Mentorship	selected from 10,000+ applicants.
Google Summer of Code	selected fellow from the top 2% of 43,765 applicants; shipped a 350-hour project.
Major League Hacking	selected for the software engineering track.
CodeChef Long Challenge II	Global Rank 1.
Open-source and private work	2000+ total pull requests merged.
Technical writing	maintains technical docs and writes on systems, Go, and web development.

OPEN SOURCE

Openswap	maintained the Rust Bitcoin swap core across wallet storage, APIs, tests, docs, and protocol ergonomics.
ThunderHub	maintained the Lightning node dashboard across channel setup, asset trading, payments, and self-hosting.
XWiki Apps	built and maintained Interactive Link Visualizer across graph search, layouts, controls, and theme support.
Banco	improved data loading, contacts, health checks, and build reliability for a Lightning banking app.
Graphology	packaged the graph library for XWiki and contributed fixes and integration feedback.
Sigma.js	reported issues across dark themes, dragging, documentation, and releases during GSoC.
XWiki Platform	fixed tagging bugs and shipped feature improvements.

PROJECTS

unixdm	 wthrajaat/unixdm
→ C++17 terminal download manager with segmented downloads, resumable state, batch queues, and an interactive TUI.	
C++17, CMake, POSIX, FTXUI, Shell	
• Built a C++17 terminal download manager with segmented downloads, batch queues, scheduling, and resume files. • Implemented worker rebalancing, retries, preallocation, and an FTXUI interface with live download feedback.	
orphanode	 wthrajaat/orphanode crates.io
→ Rust crate that finds dead code in JavaScript and TypeScript projects through reachability analysis.	
Rust, JavaScript / TypeScript, Static analysis, SARIF, Cargo	
• Written in Rust; proves code unused via a reachability graph over imports, exports, and workspace packages. • Published on crates.io and npm with prebuilt binaries, SARIF output, safe fix previews, and CI speed budgets.	
Coreloop	 wthrajaat/coreloop
→ Self-hosted learning and news platform with a Next.js control surface, durable Go services, and Telegram delivery.	
Go, Next.js / TypeScript, Turso / libSQL, Telegram Bot API, QStash, Groq / Gemini	
• Designed Go services around a Turso-backed durable queue handling scheduling, leases, retries, and provider health. • Built a Next.js control surface and Telegram delivery path with source-backed content, Groq/Gemini routing, and deterministic fallbacks.	
autocommit-rs	 wthrajaat/autocommit-rs crates.io
→ Published Rust CLI for generating conventional commit messages from staged Git changes.	
Rust, Git, OpenAI, Gemini, Cargo	
• Built and published a Rust CLI for conventional commit generation from staged Git changes. • Added interactive setup, OpenAI/Gemini support, configurable message style, signing, and hook controls.	
Interactive Link Visualizer	 xwiki-contrib/application-interactive-link-visualizer extensions.xwiki.org
→ XWiki extension for searchable page relationship graphs.	
TypeScript, XWiki, Solr, Graphology, Sigma.js	
• Integrated backlinks, titles, URLs, and document references through XWiki APIs. • Added search, facets, filters, theme-aware graph UI, documentation, and demos.	
Seamcarving	 wthrajaat/seamcarving
→ C++ content-aware image resizing with OpenCV and CMake.	
C++, OpenCV, CMake, GLEW	
• Implemented seam identification and removal with dynamic programming. • Built a C++ binary around OpenCV, GLEW, and CMake.	

EDUCATION

Bachelor of Engineering, Panjab University	Year: 2025, Grade: 7.25/10
High School, CBSE	Year: 2020, Grade: 9.56/10